

DDQN/DQN Overestimation Across Target Network Frequencies: Decoupled Q-Value Scale and Policy Performance

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Abstract

DQN and related algorithms have a well known problem of overestimation. This paper leverages a theoretical upper bound of the Cartpole environment as a proxy measure for overestimation. This paper begins by probing at DQN/DDQN overestimation differences, but found that target network update frequency (K) has a much larger impact on these violations as well as more general impact on learned Q value scale. This relation between update frequency, Q scale and learnt policy are explored in greater depth; finding that the later two appear to be strongly decoupled. The paper then establishes whether this learned Q value scale has any effect on policy. It was found that networks that exceedingly violate this upper bound or having inappropriate Q value scales can still achieve near-optimal policy whilst also effectively mapping Q values to normalised episodic discounted returns. We then offer a candidate explanation as to why, in these cases, training is still possible and why this phenomena maybe specific to survival like environments such as Cartpole; where the absence of a next frame is often a stronger learning signal than the reward for reaching the next.

1 Introduction

DQN is susceptible to overestimating the future rewards of a position and in some cases this can harm training and in turn final policy performance. DDQN is a classic fix and was shown to be effective [3] at reducing this overestimation in more complex Atari Environments. This work investigates overestimation in a simpler Cartpole environment where a theoretical upper bound is used as a proxy measure for overestimation.

Initially probing at whether DQN/DDQN had a meaningful impact on these upper bound violations, it was found that target network update frequency had a more dominant effect on this overestimation and more generally the scale of Q value predictions. This motivates the main central idea behind the report, how can networks whose Q value scales are so varied and in many instances theoretically impossible, still learn effective policies. We pursue this question by examining policy performance across Q scales as well as performance of value estimates when compared to realised episodic returns.

2 Problem Statement

Overestimation in DQN & DDQN

DQN [2] aims to map each available action in a discrete action space to the expected discounted rewards from taking that action. DQN uses an online network for policy with an offline target network for evaluation and it is well noted for DQN to overestimate its expected discounted rewards. Simply, it tends to overestimate the Q value from each actions available and this can in some situations harm training and overall policy performance.

DDQN was proposed as a variation of DQN in which the slightly older target network is only used to evaluate the online network's policy. Explicitly:

$$DQN : Q_{target} = r_t + \gamma * Q(s_{t+1}, \operatorname{argmax}(Q(s_{t+1}, a_{t+1} \mid \theta^-) \mid \theta^-) \quad (1)$$

$$DDQN : Q_{target} = r_t + \gamma * Q(s_{t+1}, \operatorname{argmax}(Q(s_{t+1}, a_{t+1} | \theta) | \theta^-)) \quad (2)$$

Where θ represents the current online model and θ^- is the slightly older 'target' network. This older target network is brought up to the current model every K time steps and this is a hyper parameter of the algorithm.

3 Experimental Set Up

3.1 Cartpole Environment

Cartpole has been used as an learning environment for two reasons.

1. It is computationally cheap
2. It has an upper bound to its maximum possible discounted reward.

The discounted reward G is a weighted sum of future rewards and is the quantity a trained DQN network wants to map to each action. In cart pole a reward of 1 is given to the agent for each frame it survives and 0 for termination.

$$\Rightarrow G_t = \sum_{k=0}^T r_{t+k} * \gamma^k \quad (3)$$

$$\Rightarrow G_t = \frac{1}{1-\gamma} \text{ as } T \rightarrow \infty \quad (4)$$

This means that even an optimal policy, that survives indefinitely, has an upper G boundary. So for our experiment, where $\gamma = 0.99$, this becomes 100. Therefore any network who predicts that the expected future discounted reward $Q > 100$ is incompatible with this fact and therefore has categorically overestimated its strength. Note, this is not the same as comparing to an underlying truth Q^* however, we will use this theoretical violation/excess as a proxy measure for a network's susceptibility to overestimation. We are defining excess as...

$$Excess(s_t, a_t, \gamma) = Q(s_t, a_t) - Upper(G(s_t, a_t)) \quad (5)$$

$$\Rightarrow Excess(s_t, a_t, \gamma) = Q(s_t, a_t) - \frac{1}{1-\gamma} \quad (6)$$

3.2 Training Regime and Network Architecture

A brief note is necessary to outline the training regime used across this report.

All training uses a small MLP network of $4 \rightarrow 128 \rightarrow 128 \rightarrow 2$ with ReLU hidden activations and a linear final layer.

Whilst training, targets Q values were set by the corresponding algorithm, either DQN or DDQN.

Each training regime was run over 10 seeds :[10 - 19] and were left to train for 100,000 time steps.

Target Network Update frequency K was initially varied on the set [50,200,500,1000] and then additionally along [535,591,647,758]

Training began after 1000 time steps and a replay buffer [2] of 25,000 was used.

Gradient updates were completed every 10 time steps with a batch size of 128 and lr = 2e-4, ADAM [1] Optimiser was used.

Policies were evaluated on a greedy basis against 25 episodes with a maximum length of 3000

3.3 Research Questions

This paper investigates a few relations:

RQ1: Does DDQN reduce the frequency/magnitude of the upper bound violations?

RQ2 : How does the target update frequency K effect violation rates, Q value scale and learned policy?

RQ3 : Is Q value accuracy correlated to Policy Performance?

We will investigate these by running 10 seeded training runs over 100000 time steps of training, checkpointing every 1000 time steps.

To evaluate Q values, each network was shown the same $\sim 100k$ feasible Cart-pole states and their Q values were plotted to establish whether the same states shown to different networks produce wildly different Q values. These 'feasible' states were generated by letting separate models of varying abilities (50-500) play episodes and recording the positions they had reached.

Similarly, each policy was evaluated by starting each checkpoint from the same 25 states, imposing an artificial 3000 episode frame limit and summing the total collected reward.

4 Findings

4.1 RQ1 - Does DDQN Reduce The Frequency/Magnitude of Theoretical Violations

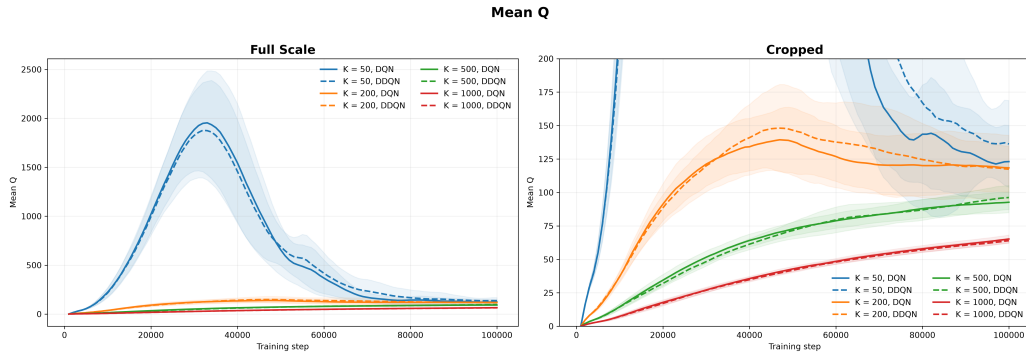


Figure 1: Evolution of Mean Q Through Training

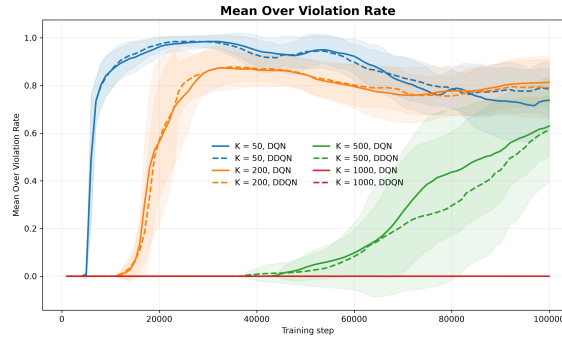


Figure 2: Mean Rate of Violations

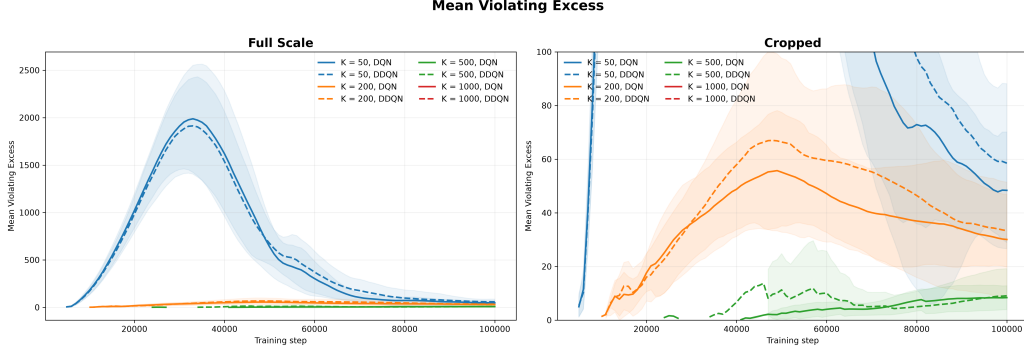


Figure 3: Mean Magnitude of Violations

The above figures show no empirical evidence for the hypothesis that DDQN reduces the rate of Q violations or even the magnitude of these violations. The dotted DDQN seems to track the DQN’s seed averaged training curve in almost every K scenario. Whereas the DDQN paper [3] had rather convincingly demonstrated that in more complex environments, DDQN had been able to significantly reduce these Q value overestimations, this report is unable to replicate this. This is expected as it was claimed that relative overestimation between DQN and DDQN methods increases with action space [3]. Therefore as our action space is 2, this difference should be at a complete minimum.

4.2 RQ2a - Does the target update frequency K effect these violations?

What is clearly demonstrated in the above figures is that K value is extremely influential on the expected overestimation in this setting[4]. It is observed that even though the rate of theoretical violations is significant for 50,200,500, the magnitude of their excess is heavily impacted. The below plots show a) policy performance across seeds and b) The difference in initial Q value prediction vs the realised episodic G value. Both of these averaged along the final 10 checkpoints, with all training seeds plotted.

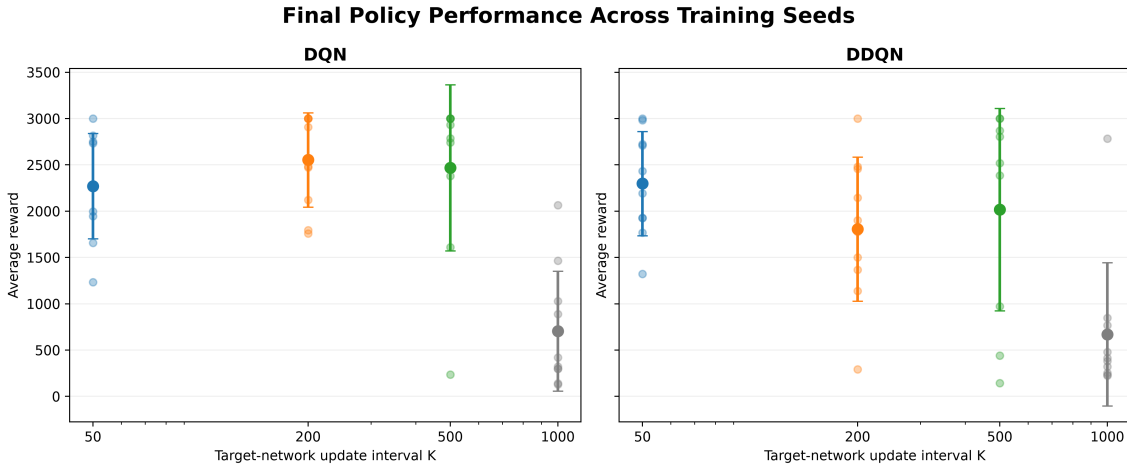


Figure 4: Policy Performance for Different K Values over Last 10 Training Steps

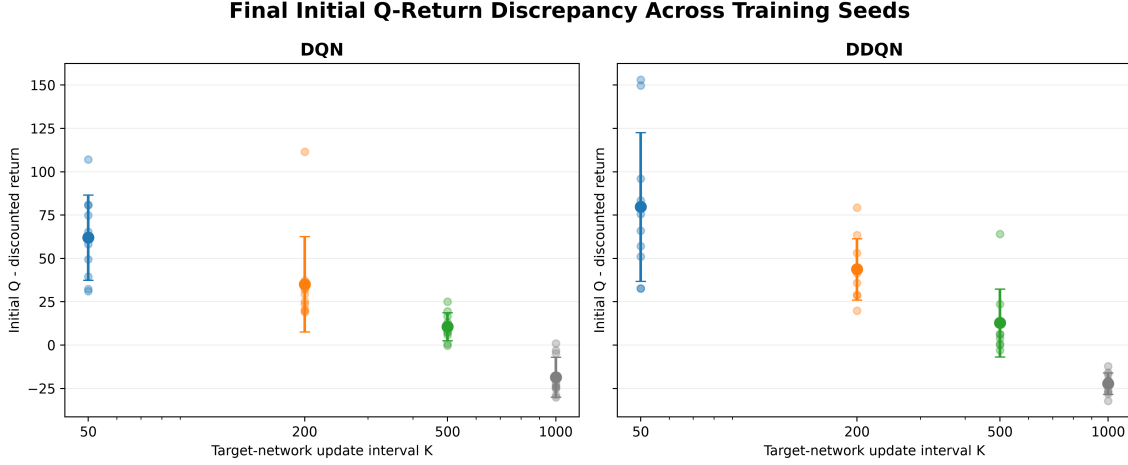


Figure 5: Initial Q and Realised G Discrepancy over Last 10 Training Steps

Now we don't claim the metric of $Q_0 - G_{realised}$ is a fair evaluation in the of accuracy of Q value predictions. At the end of the day, most episodes will be exceeding 200-300 frames and by that point G_0 has already got to ~ 100 . This means that an initial guess of 100 with a semi-effective learned policy will score you very well on this metric. This metric does however help understand the learned scale of the network's Q value predictions. The metric of $Q_0 - G_{realised}$ tracks extremely well with the rate and magnitude of Q value violations. The new question being, can we tune K such that its scale aligns with the scale of the experiment.

4.2.1 RQ2b - Can we predict a zero crossing K and if so what is the effect on the learned policy?

From simple analysis, we can try approximate the relation between $\ln(K)$ and overestimation as linear, and if one takes this as true, it seems to imply some sort of K value for which over-estimation is at a minimum. If we take only DQN runs for simplicity, for each k,seed pair average its initial excess over the last 10 steps. We can then find the seed average and fit to log linear fit giving a 0-crossing at $K_0 = 624$. A paired bootstrap over seeds gives a 95% confidence interval $530 < K_0 < 760$. As a small scale experiment the same training regime was run over 4, K values in this range{535, 591, 647, 758}. Over the same 10 seeds, the following policies were found:

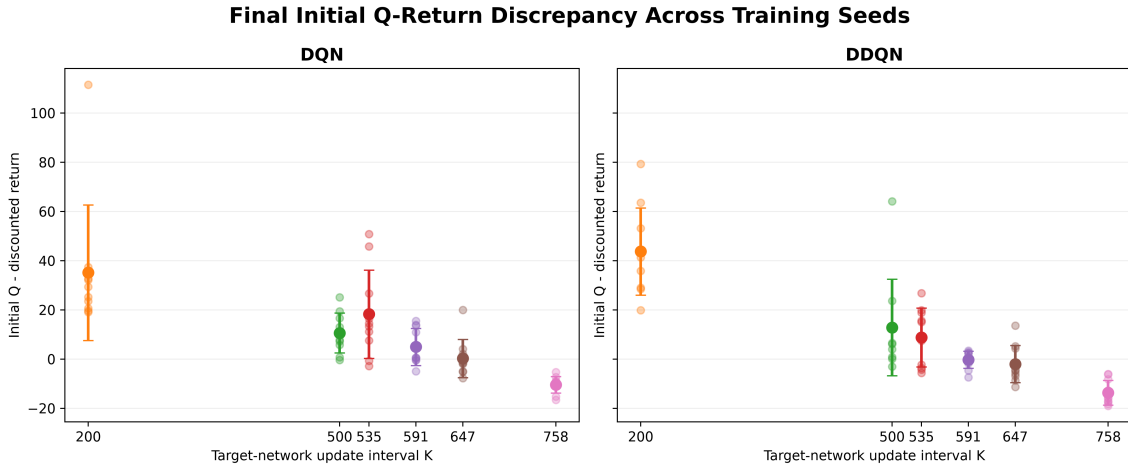


Figure 6: Initial Q vs G Discrepancy of Tuned K Values

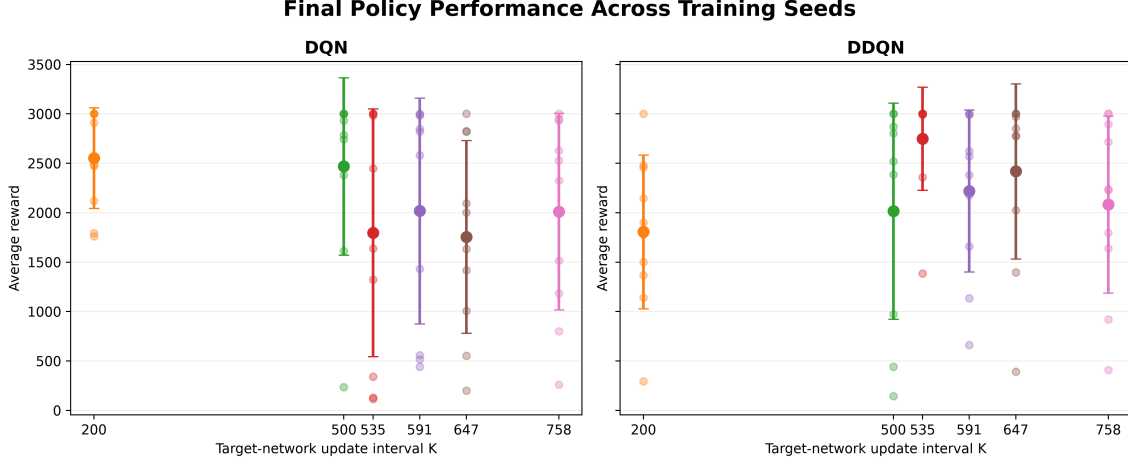


Figure 7: Policy Performance of Tuned K Values

Figure 6 shows us that this K tuning was successful in minimizing $Q_0 - G_{realised}$ and we can achieve a near 0 error for $K = 591-647$ seeds. However this results seems to be overshadowed by the fact that there is no clear evidence that this 'more accurate' Q scale has lead to any tangible improvement in policy:fig 7. This result allows us to very loosely claim a degree of separation between accuracy of Q scale and policy. In this small scale roll-out, there is no evidence to support any causal link.

4.2.2 RQ2c - Does Q value Scale Impact Correlation to Realised Discounted Reward?

The previous section (RQ2b) shows no evidence that the scale of Q values has any effect on policy and we are able to show that K can somewhat be chosen to give specific Q scales. So the question is, why bother, does the scale of the Q values learnt effect anything at all. One idea is that more fairly scaled Q values may trace realised G better, becoming more effective at evaluating the 'value' of a given state.

20 separately trained and out of sample policies were run over 10_000 of our feasible states. All of these states were set up as the initial state and all 20 policies played though the episode until termination/truncation. The total average reward for each of these starting states was recorded. States between the 10% and 20% percentiles were recorded. Put simply, 1000 poor but not completely awful positions were collected.

The last 10 training steps of all our in sample policies were rolled out against these 1000 bad states. The evolution of G and Q values throughout episodes that ended in termination were recorded and rescaled as according to (7).

$$Q_{norm} = \frac{Q - Q_{mean}}{\sigma_Q} \quad (7)$$

$$G_{norm} = \frac{G - G_{mean}}{\sigma_G} \quad (8)$$

Effectively we are tracking the Q and G though each episode that ended in policy failure and we want to see how the predicted Q moves with this $G_{realised}$

$$\rho_{spearman} = \rho(G, Q) \in [-1, 1] \quad (9)$$

$$tracking\ err = \sum_{t=0}^T \frac{|Q_{t\ norm} - G_{t\ norm}|}{T} \quad (10)$$

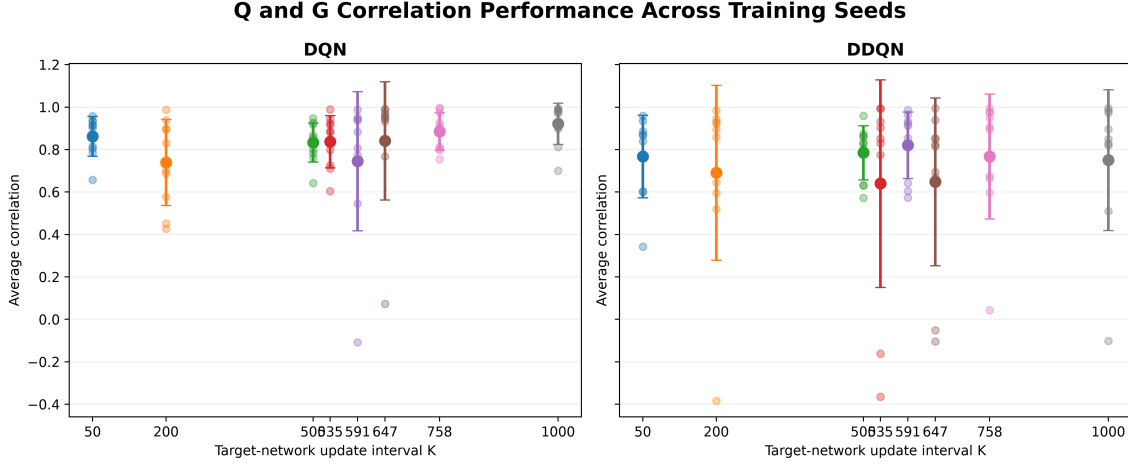


Figure 8: Terminal Episodes Q_t and G_t Correlation from 'Hard Positions'

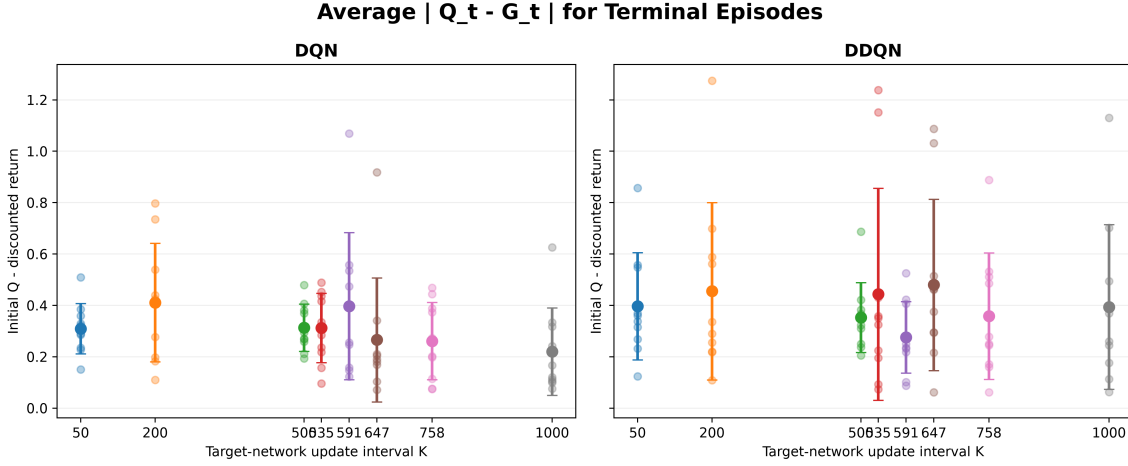


Figure 9: Average $|Q_t - G_t|$ of Terminal Episodes from 'Hard Positions'

In these charts, we are using these correlation and tracking error metrics to try and probe how well each network's normalised Q scores map to the normalised realised G along terminal trajectories. The plots again show each seed's average over its last 10 saved checkpoints. Even though Q value scales varies from large overestimation to underestimation, the apparent movement of Q and G seems to be irrelevant to the scale they occur on. Interestingly, this seems to imply that even though the Q targets are derived from the Bellman Equation, directly linking to a ground truth metric G, the policy can still sufficiently learn and function without having to adhere to this ground-truth scale and reward system.

4.3 RQ3 Is there a clear link between Q value and Policy Performance

This question has been mostly addressed up until this point, but it would be nice to lay out the evidence more clearly.

Here I will lay out the claim, that within this experiment, expected Q value scale/behaviour has little evidential correlation to policy performance.

This claim is supported though a few figures.

Fig 3&4 It is evidenced though graph 3, that the magnitude of these theoretical Q value violations moves inversely to K. Fig 4 shows small but not significant variation in policy performance across K =

50,200,500.

If it were the case that the accuracy of Q value predictions mapped onto improved policy performance, this would simply not be the case. If true, in theory we should see $K = 50$ perform the worst and $K = 1000$ the best as this had no Q violation at all. Clearly this was not the case.

Fig 5,6,7 Inspecting the fig 5 graph, which aims to probe at the scale of the Q values converged upon by the network. Using the K value tuning described, we were able to scan over K values and get small initial Q vs G errors, suggesting we had fitted the Q values to the scale of environment G values. The question is then, if this scale of Q values is matched well in $K = 591,647$ do these learned policies perform better. The claim that this was the case is weak, in DQN roll-outs, this Q value had no effect on policy, being outperformed by $K = 200$. Although the converse is true for DDQN, it is unlikely this is anything more than statistical noise. Therefore, this paper is willing to stand on a position that the accuracy and Q value scale is likely uncorrelated to performance in this small-scale experiment.

Diverging Q values in Fig 1 One of the ugliest results of the report is the exploding Q values for $K = 50$ during mid-stages of training.

We can very simply compare the policy at this stage with $K=200,500$ etc at the same training steps.



Figure 10: Early Training Policy Performance

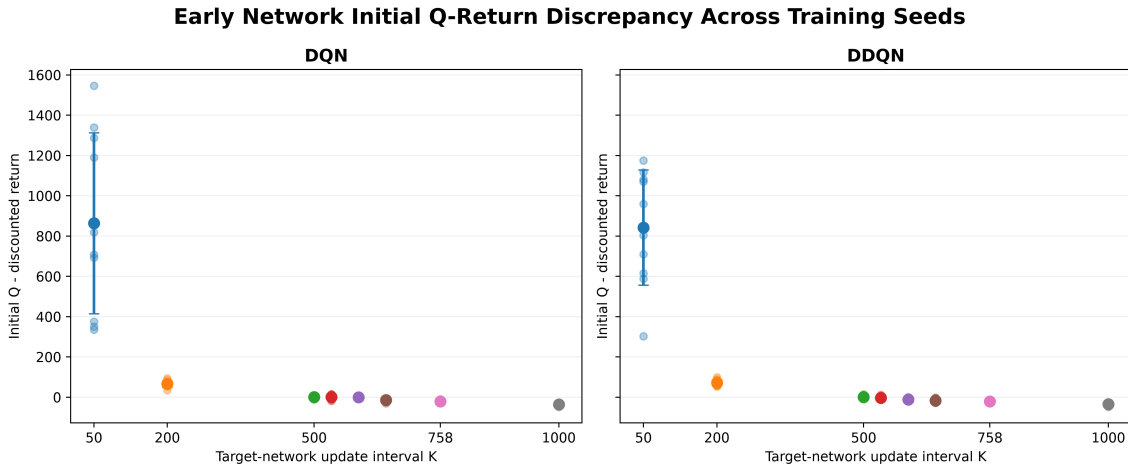


Figure 11: Early Training $Q_o - G_0$ Discrepancies

These final figures should add weight to this claim even further. Even though $K = 50$ is overestimating by an order of magnitude when compared to all other K values, its mean outperforms all but $K=200$ of which it is roughly equal to. Although some statistical nuance has to be given due to the fact that both $K = 50$ and $K = 200$ give quite high variance policies across seeds, $K = 50$ produces a larger mean and maximum policy than almost all other K values. At an absolute minimum, $K=50$ has at least produced an equivalent policy to K values which produce far 'better' Q value scales, whilst operating on a vastly inappropriate Q scale.

5 Conclusion

One thing should be adamantly clear, any claims in the report are restricted to the Cartpole Environment and similar survival oriented games. This paper has made no effort to generalise or show that these conclusions can be drawn on more complex environments. Having said that, this paper has done sufficient leg work to show three main takeaways:

1. There is no empirical evidence to show that DDQN reduces these theoretical upper bound violations.
2. K Value has an extreme impact on Overestimation Rate and Magnitude.
3. Q value scale has little - no impact on policy performance and correlation to realised G

5.1 Why does Q scale seemingly not effect learning?

This policy/ Q -scale invariance is due to fair reason in deployment, DQN uses $\text{argmax}(\{Qs\})$ so the scale of Q is irrelevant. However the reasoning as to why training doesn't collapse as Q deviates away from truth is maybe more intricate and may be limited to survival like environments. During training the Q target value is set as follows, with eq 12 following the DQN/DDQN convention set out in eq 1&2.

$$Q_{t \text{ target}} = r_{term} \quad \text{for terminal } t \quad (11)$$

$$Q_{t \text{ target}} = r_t + \gamma * Q_{algorithm \text{ dependent } t+1 \text{ max}} \quad \text{for non-terminal } t \quad (12)$$

If the previous action caused the episode to terminate, a termination reward r_{term} is incurred. In our case this r_{term} was assigned a value of 0, this is insightful. Any scale of Q as long as it is positive will get genuine learning signal from a $Q = 0$ target. Say the Q learned scale is in the order of $(1e + 5)$, a reward of 1 or 0 is nothing to the network, this is just noise. However what is actual signal to the network is the second half of eq. 1 & 2. Missing out on a $\gamma * (1e + 5)$ order of magnitude 'reward' through termination is the real punishment. Therefore it is this paper's hypothesis that these conclusions are only applicable to this structure of environment in which termination is a possibility and avoiding termination is the goal of the policy.

For example, lets take the same Cartpole environment but it no longer terminates if the pendulum/cart deviates too far from the centre and but instead exists for a fixed episode length. Instead we reward it on a reward scale $[0,1]$ where a upward vertical pole is 1 and a downward vertical is 0, and everything in-between. It is this paper's hypothesis, in this type of environment, policy performance is likely to be far more sensitive to Q as it needs to be on an appropriate scale to get impactful learning signal from these small/scale dependent rewards.

The clear next steps is investigating this claim performing similar experiments to evaluate whether this conclusion is fair.

References

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